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Residuals

Graded Quiz • 30 min

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Introduction

Residuals

Quiz

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Residuals

Review Learning Objectives

1. Throughout this quiz, $Y \sim N(X\beta, \sigma^2 I)$. $H_X = X(X^T X)^{-1} X^T$, h_{ij} is element (i, j) of H_X and x_i is row i of X expressed as a column vector. The number h_{ii} is equal to

Correct

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Due Mar 14, 11:59 PM IST Attempts 3 every 8 hours

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2. Let S be a subset of the integers $1, \dots, n$. Let X_{-S} and Y_{-S} be the design matrix and response vector with those rows removed. Let x_i ($p \times 1$) and y_i (scalar) be the rows of X and elements of y . What is equivalent to $X_{-S}^T X_{-S}$?

Correct

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3. Let S be a subset of the integers $1, \dots, n$. Let X_{-S} and Y_{-S} be the design matrix and response vector with those rows removed. Let x_i ($p \times 1$) and y_i (scalar) be the rows of X and elements of y . What is equivalent to $X_{-S}^T Y_{-S}$? (Check all that apply.)

Correct

4. Let h_{ij} be element i, j of the hat matrix. The standardized residual $e_i / \sqrt{(1 - h_{ii}) S^2}$ follows a T-distribution.

Correct

5. Under the standard linear model normality assumptions, the residuals are independent from $\hat{\beta}$?

Correct

6. Under the standard normality assumptions, the residuals are independent from $\hat{Y}$?

Correct

7. The hat diagonals h_{ii} must be between -1 and 1?

Correct

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1/1